

MTH313M PRACTICE PROBLEMS WEEK 6 AND 7

Question 1. Consider a nonhomogeneous Poisson Process $\{N_t\}_{t \geq 0}$ with rate function $\lambda(t) > 0, \forall t$. Recall $m(t) = \int_0^t \lambda(s) ds$. Define $\{M_t\}_{t \geq 0}$ as $M_t = N_{m^{-1}(t)}$. Identify the distribution of $\{M_t\}_{t \geq 0}$.

Question 2. Consider an M/M/1 queueing system with the queue discipline following last-come first-served policy. Let X_t denote the number of customers in the system at time $t \geq 0$. Show that $\{X_t\}_{t \geq 0}$ is a Birth & Death Process.

Question 3. Let $\{B_t\}_{t \geq 0}$ be a standard Brownian motion. Find the joint distribution of $\begin{pmatrix} B_1 \\ B_3 + B_5 \\ B_9 \end{pmatrix}$